

Flavor Mixing from Boundary Misalignment in Hyperbolic Geometry

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Abstract

We formalize a kinematic mechanism by which fermion mixing matrices arise from the misalignment between discrete boundary sectors associated with logarithmic mass organization in hyperbolic space. Mixing matrices are defined as overlap operators between preferred orthonormal bases of sector subspaces and are shown to be uniquely determined up to sector-local rephasings. Our main result (Theorem 7.2) establishes that the set of accessible mixing matrices is discrete in $U(n)$ modulo sector-local rephasings, providing a structural explanation for the stability of mixing patterns without continuous parameter tuning.

1 Introduction

The Standard Model fermion mixing matrices exhibit stable, non-generic patterns unexplained by gauge structure alone. We develop a kinematic explanation: mixing arises from the misalignment between boundary sectors naturally associated with discrete logarithmic mass organization.

If fermion mass scales are organized near a discrete logarithmic lattice, each fermion is associated with a definite direction on the boundary sphere $\partial\mathbb{H}^3 \cong \mathbb{S}^2$. Different generations of a given sector span different subspaces of a boundary Hilbert space $\mathcal{H}$; their overlap is the mixing matrix. The key structural result is that if boundary directions are assigned by a *discrete* selection rule, the set of accessible mixing matrices is itself discrete (Theorem 7.2).

2 Geometric Setup and the Discreteness Assumption

Definition 2.1 (Boundary direction). A *boundary direction* is a point $\xi \in \partial\mathbb{H}^3 \cong \mathbb{S}^2$. Group elements $\gamma \in \Gamma$ are *associated with* ξ when the geodesic ray from basepoint 0 through $\gamma \cdot 0$ converges to ξ .

Assumption 2.2 (Discrete sector assignment). Each fermion flavor $f \in F$ is assigned a unique boundary direction $\xi(f) \in \partial\mathbb{H}^3$ by a deterministic, discrete selection rule: the image $\Xi = \{\xi(f) : f \in F\}$ is finite, and the assignment is locally rigid. One concrete realization: $\xi(f)$ is the ideal endpoint of the geodesic ray to the minimal representative $\gamma_f \cdot 0$, whose uniqueness and rigidity follow from the companion paper on operator realization [1].

3 Boundary Hilbert Space and Localized States

Definition 3.1 (Boundary-localized state). For each $\xi \in \mathbb{S}^2$, fix a normalized state $|\xi\rangle \in \mathcal{H}$ with $\langle \xi | \xi \rangle = 1$, depending continuously on ξ , with $\langle \xi | \zeta \rangle = 1$ iff $\xi = \zeta$.

4 Sector Subspaces and Preferred Bases

Definition 4.1 (Sector subspace). The *sector subspace* of S is $\mathcal{H}_S = \overline{\text{span}}\{|\xi(f)\rangle : f \in S\} \subset \mathcal{H}$, with $\dim \mathcal{H}_S = |S|$ by linear independence.

Definition 4.2 (Preferred orthonormal basis). The *preferred ONB* of $\mathcal{H}_S$ is the Gram–Schmidt orthonormalization of $(|\xi(f_1)\rangle, \dots, |\xi(f_{n_S})\rangle)$ in the fixed sector ordering, with all pivot coefficients real and positive.

5 Mixing Matrices as Overlap Operators

Definition 5.1 (Mixing matrix). For sectors S, T of equal cardinality n with preferred ONBs $\{|S, i\rangle\}$ and $\{|T, j\rangle\}$, the *mixing matrix* is $(U_{ST})_{ji} = \langle T, j | S, i \rangle$.

Proposition 5.2 (Unitarity). If $\mathcal{H}_S = \mathcal{H}_T$, then $U_{ST} \in \text{U}(n)$.

Theorem 5.3 (Geometric dependence). U_{ST} is determined entirely by $\Xi_{ST} = \{\xi(f) : f \in S \cup T\}$ and the inner product on $\mathcal{H}$.

6 Rephasing Invariance

Proposition 6.1 (Rephasing-invariant quantities). Under sector-local rephasings $|S, i\rangle \mapsto e^{i\alpha_i} |S, i\rangle$, $|T, j\rangle \mapsto e^{i\beta_j} |T, j\rangle$, the following are invariant: (i) the moduli $|(U_{ST})_{ji}|^2$; (ii) the quartic products $J_{ji,j'v} = (U_{ST})_{ji}(U_{ST})_{j'v}\overline{(U_{ST})_{ji'}}\overline{(U_{ST})_{j'i}}$; (iii) $\det(U_{ST})$ up to overall phase.

7 Discreteness and Stability of Mixing Classes

Theorem 7.1 (Stability). For $\varepsilon \leq \varepsilon_0 = \delta_{\min}/2$, where $\delta_{\min} = \min_{f \neq f'} d_{\mathbb{S}}^2(\xi(f), \xi(f'))$, any bounded deformation of magnitude ε leaves Ξ , the preferred bases, and U_{ST} unchanged.

Theorem 7.2 (Discreteness of mixing classes). Under Assumption 2.2, the set of accessible mixing matrices $\mathcal{M}_{ST}$ is discrete in $\text{U}(n)/(\text{U}(1)^n \times \text{U}(1)^n)$.

Proof. The set of admissible configurations Ξ_{ST} is finite (discrete). By Theorem 5.3, the map $\Xi_{ST} \mapsto U_{ST}$ is continuous, so it maps a finite set to a finite set. Quotienting by rephasings preserves finiteness. $\square$

Corollary 7.3 (No continuous drift). No continuous deformation of physical parameters can produce continuous drift in any rephasing-invariant mixing observable.

8 Conclusion

We have shown that fermion mixing matrices arising from boundary-sector misalignment are unitary by construction, rephasing-invariant, stable under bounded deformations, and discrete modulo rephasings. These properties are structural consequences of the discrete selection rule and require no dynamical input or continuous tuning.

References

- [1] M. R. Bridson and A. Haefliger. *Metric Spaces of Non-Positive Curvature*. Springer, 1999.